LAB 1:

Aim:To perform and understand fundamental DBMS operations, including database creation, table management, data manipulation, modification, and retrieval using MySQL.

SHOW databases;

CREATE DATABASE stud1_test;

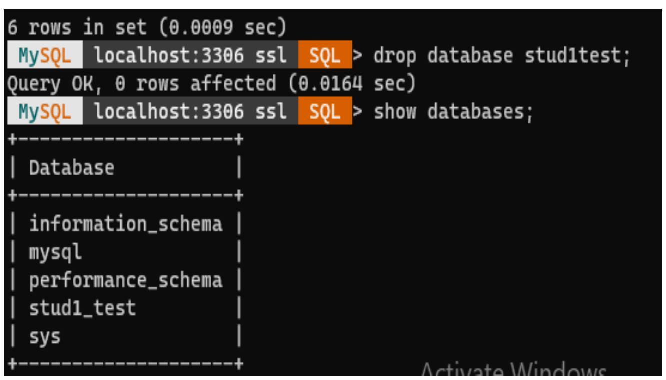
SHOW DATABASES;

CREATE DATABASE stud1test;

SHOW DATABASES;

DROP DATABASE stud1test;

SHOW databases;



USE stud1_test;

CREATE TABLE Persons (

PersonID int,

LastName varchar(25),

FirstName varchar(25),

Address varchar(100),

City varchar(25)

);

```
 MySQL  localhost:3306 ssl  stud1_test  SQL > show tables;
+----------------------+
| Tables_in_stud1_test |
+----------------------+
| customers            |
| persons              |
+----------------------+
2 rows in set (0.0013 sec)
 MySQL  localhost:3306 ssl  stud1_test  SQL > describe Persons;
+-----------+--------------+------+-----+---------+-------+
| Field     | Type         | Null | Key | Default | Extra |
+-----------+--------------+------+-----+---------+-------+
| PersonID  | int          | YES  |     | NULL    |       |
| LastName  | varchar(25)  | YES  |     | NULL    |       |
| FirstName | varchar(25)  | YES  |     | NULL    |       |
| Address   | varchar(100) | YES  |     | NULL    |       |
| City      | varchar(25)  | YES  |     | NULL    |       |
+-----------+--------------+------+-----+---------+-------+
```

CREATE TABLE Customers (CustomerID int,LastName varchar(25),FirstName varchar(25),Address varchar(100),City varchar(25), primary key (CustomerID));

DESCRIBE Customers;

```
 MySQL  localhost:3306 ssl  stud1_test  SQL > describ
+------------+--------------+------+-----+---------+-
| Field      | Type         | Null | Key | Default |
+------------+--------------+------+-----+---------+-
| CustomerID | int          | NO   | PRI | NULL    |
| LastName   | varchar(25)  | YES  |     | NULL    |
| FirstName  | varchar(25)  | YES  |     | NULL    |
| Address    | varchar(100) | YES  |     | NULL    |
| City       | varchar(25)  | YES  |     | NULL    |
+------------+--------------+------+-----+---------+-
5 rows in set (0.0014 sec)
```

INSERT INTO Customers (CustomerID,LastName,FirstName,Address,City)

VALUES (96365,'Shah','Hiren','A67,Building10,abc lane','Mumbai');

SELECT * FROM Customers;

```
 MySQL  localhost:3306 ssl  stud1_test  SQL > SELECT * FROM Customer
s;
+------------+----------+-----------+-------------------------+------
---+
| CustomerID | LastName | FirstName | Address                 | City
   |
+------------+----------+-----------+-------------------------+------
---+
|      96365 | Shah     | Hiren     | A67,Building10,abc lane | Mumb
ai |
+------------+----------+-----------+-------------------------+------
---+
```


INSERT INTO Customers VALUES

(96300,'Singh','Prakesh','B7,House10,abc lane','Mumbai'),

(89790,'Joshi','Ramesh','413,Sadashiv Nagar','Pune');

----Now add approx. 10 customers------

```
1 row in set (0.0005 sec)
 MySQL  localhost:3306 ssl  stud1_test  SQL > INSERT INTO Customers
VALUES
                                          -> (96300,'Singh','Prakes
h','B7,House10,abc lane','Mumbai'),
                                          -> (89790,'Joshi','Ramesh
','413,Sadashiv Nagar','Pune');
Query OK, 2 rows affected (0.0056 sec)

Records: 2  Duplicates: 0  Warnings: 0
 MySQL  localhost:3306 ssl  stud1_test  SQL > INSERT INTO Customers
VALUES
                                          -> (96301,'Patel','Aarav'
,'A12 Building, MG Road','Mumbai'),
                                          -> (96302,'Sharma','Anany
a','B45 House, FC Road','Pune'),
                                          -> (96303,'Mehta','Rohan'
,'C21 Building, Linking Road','Mumbai'),
                                          -> (96304,'Desai','Ishita
','D18 House, Station Road','Nashik'),
                                          -> (96305,'Kulkarni','Adi
tya','E32 Building, JM Road','Pune'),
                                          -> (96306,'Gupta','Priya'
,'F09 House, LBS Road','Thane'),
                                          -> (96307,'Joshi','Rahul'
,'G15 Building, Camp Area','Pune');
Query OK, 7 rows affected (0.0087 sec)
```

SELECT * FROM Customers;

```
Records: 7  Duplicates: 0  Warnings: 0
 MySQL  localhost:3306 ssl  stud1_test  SQL > SELECT *FROM Customers;
+------------+----------+-----------+----------------------------+--------+
| CustomerID | LastName | FirstName | Address                    | City   |
+------------+----------+-----------+----------------------------+--------+
|      89790 | Joshi    | Ramesh    | 413,Sadashiv Nagar         | Pune   |
|      96300 | Singh    | Prakesh   | B7,House10,abc lane        | Mumbai |
|      96301 | Patel    | Aarav     | A12 Building, MG Road      | Mumbai |
|      96302 | Sharma   | Ananya    | B45 House, FC Road         | Pune   |
|      96303 | Mehta    | Rohan     | C21 Building, Linking Road | Mumbai |
|      96304 | Desai    | Ishita    | D18 House, Station Road    | Nashik |
|      96305 | Kulkarni | Aditya    | E32 Building, JM Road      | Pune   |
|      96306 | Gupta    | Priya     | F09 House, LBS Road        | Thane  |
|      96307 | Joshi    | Rahul     | G15 Building, Camp Area    | Pune   |
|      96365 | Shah     | Hiren     | A67,Building10,abc lane    | Mumbai |
+------------+----------+-----------+----------------------------+--------+
```

ALTER TABLE Customers ADD Email varchar(50);

DESCRIBE Customers;

```
10 rows in set (0.0005 sec)
 MySQL  localhost:3306 ssl  stud1_test  SQL > ALTER TABLE Customers ADD Email varchar(50);
Query OK, 0 rows affected (0.0188 sec)

Records: 0  Duplicates: 0  Warnings: 0
 MySQL  localhost:3306 ssl  stud1_test  SQL > DESCRIBE Customers;
+------------+--------------+------+-----+---------+-------+
| Field      | Type         | Null | Key | Default | Extra |
+------------+--------------+------+-----+---------+-------+
| CustomerID | int          | NO   | PRI | NULL    |       |
| LastName   | varchar(25)  | YES  |     | NULL    |       |
| FirstName  | varchar(25)  | YES  |     | NULL    |       |
| Address    | varchar(100) | YES  |     | NULL    |       |
| City       | varchar(25)  | YES  |     | NULL    |       |
| Email      | varchar(50)  | YES  |     | NULL    |       |
+------------+--------------+------+-----+---------+-------+
```

ALTER TABLE Customers ADD Remark varchar(255);

DESCRIBE Customers;

```
Records: 0  Duplicates: 0  Warnings: 0
 MySQL  localhost:3306 ssl  stud1_test  SQL > DESCRIBE Customers;
+------------+--------------+------+-----+---------+-------+
| Field      | Type         | Null | Key | Default | Extra |
+------------+--------------+------+-----+---------+-------+
| CustomerID | int          | NO   | PRI | NULL    |       |
| LastName   | varchar(25)  | YES  |     | NULL    |       |
| FirstName  | varchar(25)  | YES  |     | NULL    |       |
| Address    | varchar(100) | YES  |     | NULL    |       |
| City       | varchar(25)  | YES  |     | NULL    |       |
| Email      | varchar(50)  | YES  |     | NULL    |       |
| Remark     | varchar(255) | YES  |     | NULL    |       |
+------------+--------------+------+-----+---------+-------+
7 rows in set (0.0014 sec)
```

SELECT CustomerID,Address FROM Customers;

```
Rows matched: 1  Changed: 1  Warnings: 0
 MySQL  localhost:3306 ssl  stud1_test  SQL > SELECT CustomerID,Address FROM Customers;
+------------+---------------------------+
| CustomerID | Address                   |
+------------+---------------------------+
|      89790 | 413,Sadashiv Nagar        |
|      96300 | B7,House10,abc lane       |
|      96301 | A12 Building, MG Road     |
|      96302 | B45 House, FC Road        |
|      96303 | C21 Building, Linking Road|
|      96304 | D18 House, Station Road   |
|      96305 | E32 Building, JM Road     |
|      96306 | F09 House, LBS Road       |
|      96307 | G15 Building, Camp Area   |
|      96365 | A67,Building10,abc lane   |
+------------+---------------------------+
```

1. What is a database? Why do we need a Database Management System such as MySQL?

A database is an organized collection of data that can be stored, accessed, and managed efficiently.

A Database Management System (DBMS) is software used to create, store, retrieve, update, and manage data in a database. MySQL is an example of a DBMS.

We need a DBMS because it helps us:

- Store large amounts of data systematically.
- Search and retrieve data quickly.
- Add, modify, and delete data easily.
- Reduce unnecessary duplication of data.
- Improve data security.
- Maintain the accuracy and consistency of data.

2. What is the difference between CREATE DATABASE, SHOW DATABASES, and DROP DATABASE?

Command	Purpose
CREATE DATABASE	Creates a new database.
SHOW DATABASES	Displays the databases available in MySQL.
DROP DATABASE	Deletes an existing database and all its contents.

Examples:

CREATE DATABASE School;
SHOW DATABASES;
DROP DATABASE School;

3. What is a table in a database? Explain the purpose of rows and columns in a table.

A table is a structured collection of related data in a database. A table consists of rows and columns.

- Columns represent the attributes or fields of the data.
- Rows represent individual records.

For example:

CustomerID	Name	Age
101	Ravi	25
102	Priya	30

In this table, CustomerID, Name, and Age are columns, while each customer is represented by a row.

4. What is a primary key? Why was CustomerID selected as the primary key in the Customers table?

A primary key is a column, or combination of columns, that uniquely identifies each record in a table.

A primary key must contain unique values and cannot contain NULL values.

CustomerID was selected as the primary key because every customer can have a unique ID. It helps identify each customer accurately, even if two customers have the same name.

For example:

CustomerID	Name
101	Ravi
102	Ravi
103	Priya

Although two customers have the same name, their CustomerID values are different.

5. What is the purpose of the VARCHAR and INT data types? Give one suitable example of each?

VARCHAR

VARCHAR is used to store variable-length text or character data.

Example:

Name VARCHAR(50)

It can store values such as "Rahul" or "Priya Sharma".

INT

INT is used to store whole numbers (integers).

Example:

CustomerID INT

It can store values such as 101, 102, or 500.

6. What is the difference between INSERT, SELECT, UPDATE, and ALTER TABLE?

Command	Purpose
INSERT	Adds new records to a table.
SELECT	Retrieves or displays data from a table.
UPDATE	Modifies existing records in a table.
ALTER TABLE	Changes the structure of a table, such as adding or modifying columns.

Examples

INSERT:

INSERT INTO Customers
VALUES (101, 'Ravi');

Adds a new customer to the table.

SELECT:

SELECT * FROM Customers;

Displays the data from the Customers table.

UPDATE:

UPDATE Customers
SET Name = 'Rahul'
WHERE CustomerID = 101;

Changes the name of customer 101.

ALTER TABLE:

ALTER TABLE Customers
ADD Email VARCHAR(100);

Adds a new Email column to the Customers table.

7. Why is the WHERE clause important in an UPDATE statement? What could happen if we write an UPDATE statement without a WHERE condition?

The WHERE clause is important because it specifies which records should be updated.

For example:

UPDATE Customers
SET Name = 'Rahul'
WHERE CustomerID = 101;

Only the customer whose CustomerID is 101 will be updated.

However, if we write:

UPDATE Customers
SET Name = 'Rahul';

without a WHERE condition, all records in the Customers table will be updated.

Therefore, we should use a suitable WHERE condition in an UPDATE statement to prevent accidentally changing all the records in a table.